Rush Hour: Bike-Sharing Demand - Project 1 report

Machine Learning, Winter 2026 | Team IDs 16002932, 16004208, 16006616 | team seed 42648. Every number below is printed by **bike_sharing_pipeline.ipynb** (one notebook, five chained phases plus the test predictions, run top to bottom).

Setup used in every phase

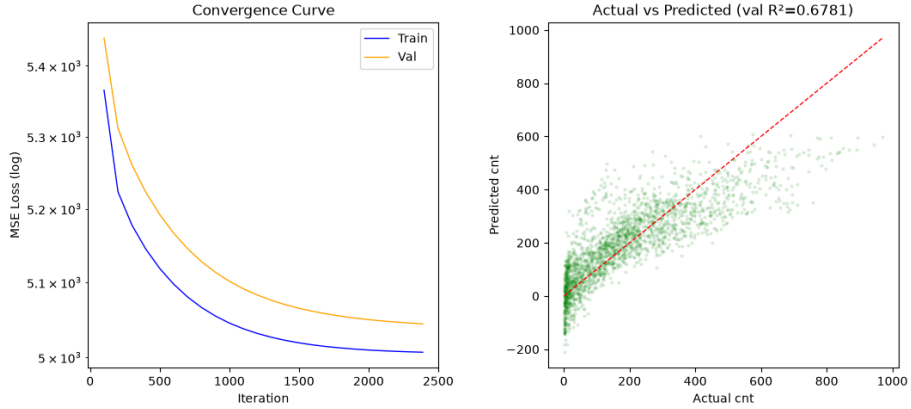
Data and split. Only *train.csv* (10,886 hourly rows, days 1-19 of every month), *test.csv* (574 rows, the 20th of every month, no labels) and *sample_submission.csv* are used. The seed is sha256 of the sorted IDs joined by '_', modulo 100,000 = **42648** (the PDF's example IDs give 41698 in our code). One seeded 80/20 split gives 8,708 training and 2,178 validation rows. Scalers, encoders, polynomial expansions and feature selection are fitted on training rows only; *test.csv* is only transformed. **Chaining.** Each phase stores its weights and hyper-parameters in memory together with the SHA-256 of the weights; the next phase re-checks that hash before it starts from them. **No sklearn model is fitted** (sklearn is used for the split, GroupKFold and metric cross-checks only); a cell of the notebook inspects every executed cell for *.fit* / *.predict* calls. Phases 3-5 and the final fit use the Phase 1 gradient descent with an optional Nesterov momentum that restarts when the loss rises, because plain gradient descent needed 15,000-150,000 steps on the 589-feature models; with it the whole notebook runs in minutes instead of 90.

Phase 1 - Gradient descent

Audit and features. windspeed is exactly 0 in 1,313 rows (12.1%, clustered at night): a missing-value code, so we add an *is_windspeed_zero* flag. hum is 0 on 22 rows of 2011-03-10 (a sensor outage, kept and noted); weathersit = 4 occurs once and is merged into 3; temp and atemp correlate at 0.985; 42 of the 456 training days have fewer than 24 hourly rows (58 missing hours). Demand peaks at 8 a.m. and 5 p.m., so *hr* cannot be one number: hr, season, mnth, weekday and weathersit are one-hot encoded (first level dropped), dteday becomes *days_since_start* (a trend that carries the 2011 to 2012 growth), and temp, atemp, hum, windspeed and days_since_start are standardised with training statistics: **54 features**. The matrix has rank 50 of 54 (season is a function of the month, workingday of weekday and holiday; condition number 1.9e16), so single weights are not trustworthy while predictions are.

Learning rate and stopping. For the loss ½·mean(residual²), gradient descent is stable only for lr < 2/L, where L is the largest eigenvalue of X^TX/n (with the intercept): L = 2.4219, so 2/L = 0.826. The sweep confirms the theory: rates from 1e-4 to 0.1 are far too slow (not converged after 20,000 iterations), 1/L, 1.5/L and 1.9/L converge, 2/L oscillates (loss 25,620, validation R^2 -0.61) and 2.1/L diverges (4e10 after 76 iterations). We chose the largest converged rate not above 1.5/L, **lr = 0.619**: fast and well inside the stable region. Stopping rule: relative loss change < 1e-6 together with gradient norm < 1 (or norm < 1e-8); divergence is declared when the loss exceeds 1e6 times its start. Result: **2,389 iterations**.

Evidence and result. Finite-difference gradient checks pass (relative error 1.6e-6 on real data), gradient descent recovers a synthetic noiseless model to 4e-9, and the loss decreases monotonically at 1/L. Against closed-form least squares the training loss differs by 1.7e-6 (relative) and validation R^2 is 0.678543 versus 0.678527. Final model: train R^2 0.6982, **validation R^2 0.6781, RMSE 100.44** (gap 0.020): it underfits, as expected from a straight line through a double-peaked daily pattern. 9.55% of the predictions are negative (clipping at 0 would give 0.6966); we leave them in the chain and floor only the final test predictions. The largest weights belong to hr_17, hr_18 and hr_8, the commute hours.



Phase 1: convergence of the final fit (left) and actual versus predicted validation counts (right).

Bonus - asymmetric loss. With e = y - ŷ and c = 3 if e > 0 (under-prediction) else 1, J = mean(c·e²), so $\partial J/\partial w = -(2/n)\cdot X^T(c\cdot e)$ and $\partial J/\partial b = -(2/n)\cdot \Sigma c\cdot e$ (differentiable at e = 0 because both quadratic pieces have zero slope there). The Hessian is at most 6 times the MSE one, so the step is 1/(6L) = 0.0688.

Compared with the MSE model on validation, the share of under-predicted hours falls from 46.69% to 32.69%, the mean residual moves from -1.80 to -43.36 (the fit sits above the data on purpose), the asymmetric cost drops from 20,602 to 17,448, and R^2 falls from 0.6781 to 0.5961.

Phase 2 - Polynomial regression

What to expand. Only the continuous columns temp, hum, windspeed and days_since_start: powers of 0/1 dummies are copies of themselves, and atemp is almost temp (adding its powers lowers validation R^2 from 0.6860 to 0.6854). Curvature evidence is weak: Pearson and Spearman correlations with cnt are close (temp 0.399/0.415, hum -0.319/-0.360) and single-feature quadratics change validation R^2 by -0.001 (temp, hum), +0.003 (windspeed) and +0.004 (days). The Phase 1 residuals are instead strongly structured by hour x working day (mean residual -277 at 8 a.m. on non-working days, +89 on working days), which no power of a weather variable can fix: our hypothesis for Phase 3.

Initialisation from Phase 1. The expanded vector is [Phase 1 weights, 0 ... 0] with the Phase 1 bias, so the predictions at iteration 0 equal Phase 1's (max difference 2e-13) and the starting loss equals Phase 1's final loss. At degree 2 the warm start converges in **178 iterations against 3,603 from zero**, to the same fit (validation R^2 0.6796 vs 0.6792). The step is 0.9/L_d (L_d = 3.30 at degree 3); the sweep again diverges above 2/L_d.

Degree. Degrees 1 to 6 give validation R^2 0.6781, 0.6796, 0.6858, 0.6865, 0.6864, 0.6866 (train 0.698 to 0.706; the gap stays about 0.02: no overfitting, simply no gain). Degrees 5 and 6 are flagged optimisation-limited (condition number 1.3e4 and 2e5, gradient descent more than 0.2% above the closed-form loss). Rule: the smallest degree whose validation R^2 is within 0.001 of the best non-limited degree (4: 0.6865). **Degree 3, 62 features: validation R^2 0.6858, RMSE 99.24** (+0.0077 over Phase 1). Cross terms between the four continuous columns would reach 0.7146 (closed form), still far below what hour interactions achieve in Phase 3.

Phase 3 - Bias-variance tradeoff

What we varied. Powers of weather columns barely move R^2 (Phase 2), so the relevant complexity axis is interaction structure. Anchored on Phase 2's degree 3 we built a ladder of 12 nested models: R0 linear (Phase 1), R1 Phase 2's model, R2 weather powers only (drops the powers of days_since_start), then R3-R11 adding one interaction family at a time: hr x workingday, hr x weather, hr x weekday, hr x mnth, hr x temp, hr x hum, hr x yr, workingday x yr and x mnth, then hr x windspeed, season, holiday. Each fitted rung starts from the previous rung's weights copied by feature name. Measured on every rung: train and seeded-validation R^2; **day-grouped 5-fold CV inside the training part** (whole days stay together; this, not the validation set, drives the choice); and one **chronological split** (sorted by date and hour, first 80% train, cut after 2012-08-05; no seed).

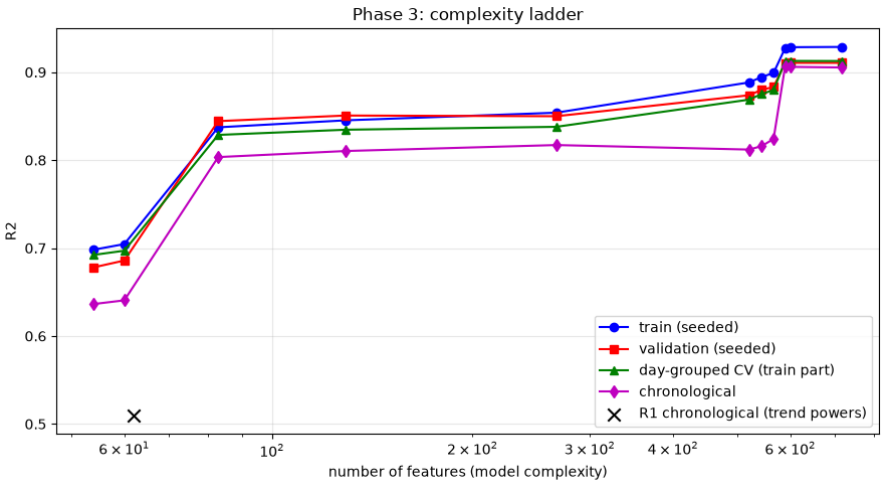
Diagnosis rule, fixed before running. High bias: R^2 low, train-validation gap < 0.02, and the next rung still raises CV R^2 by more than 0.005. High variance: gap > 0.04, or CV/chronological R^2 falls when capacity is added. Carry forward the least complex rung whose CV R^2 is within 0.005 of the best (0.9128, threshold 0.9078).

rung	features	train R^2	validation R^2	gap	CV R^2	CV std	chronological R^2
R0 linear (Phase 1)	54	0.6982	0.6781	0.0200	0.6923	0.0132	0.6364
R1 Phase 2 model (+ days powers)	62	0.7044	0.6858	0.0186	0.6969	0.0103	0.5099
R2 weather powers only	60	0.7045	0.6859	0.0187	0.6970	0.0103	0.6406
R3 + hr x workingday	83	0.8373	0.8444	-0.0070	0.8287	0.0101	0.8035
R4 + hr x weather	129	0.8453	0.8507	-0.0055	0.8346	0.0069	0.8104
R5 + hr x weekday	267	0.8540	0.8499	0.0040	0.8379	0.0080	0.8172
R6 + hr x mnth	520	0.8884	0.8738	0.0146	0.8688	0.0060	0.8120
R7 + hr x temp	543	0.8945	0.8799	0.0145	0.8752	0.0082	0.8159
R8 + hr x hum	566	0.8992	0.8832	0.0160	0.8797	0.0087	0.8241
R9 + hr x yr (carried forward)	**589**	**0.9273**	**0.9104**	**0.0170**	**0.9125**	**0.0058**	**0.9063**
R10 + workingday x yr, x mnth	601	0.9284	0.9107	0.0178	0.9128	0.0054	0.9061
R11 + hr x windspeed, season, holiday	716	0.9287	0.9106	0.0181	0.9127	0.0053	0.9054

Diagnosis: bias, not variance. The largest train-validation gap on the ladder is 0.020 (R0; 0.017 at R9), far below 0.04, and capacity keeps paying (CV +0.132 at R3, +0.031 at R6, +0.033 at R9) until it plateaus after R9 (+0.0003, -0.0001): validation R^2 rises from 0.6858 to 0.9104. **Powers of days_since_start are a variance problem that only the time-ordered split can see**: removing them (R1 to R2) changes validation by -0.0001 and CV by +0.0001 but the chronological R^2 by +0.131 (0.5099 to 0.6406); at R9, keeping them gives chronological 0.8508 instead of 0.9063. Growth over time is carried instead by hr x yr. **Carried forward: R9, 589 features** (train 0.9273, validation 0.9104 with RMSE 53.00, CV 0.9125 ± 0.0058, chronological 0.9063; 1,934 iterations; the closed form agrees to six

decimals, 0.910371 vs 0.910370; the learning-curve gap is +0.4156 with 10% of the data and +0.0170 with all of it).

The time question. For the carried model the seeded (0.9104) and chronological (0.9063) estimates differ by 0.004, whereas for Phase 2's model they differed by 0.176 (0.6858 vs 0.5099): the random split lets the model interpolate between hours of the same day and hides the extrapolation failure, so for genuinely new periods the chronological estimate is the one to read, and the diagnosis of Phase 2's model changes from 'merely too simple' to 'also fragile out of time'. *test.csv* is, however, the 20th of every month and lies inside the training period, so the day-grouped CV (0.9125) and a holdout of days 18-19 of every month (0.8967) are the closest proxies for it; all four estimates lie between 0.897 and 0.913.



Phase 3: train, validation, day-grouped CV and chronological R² along the complexity ladder (black cross: Phase 2's model on the chronological split).

Phase 4 - Regularization (L1, L2, Elastic Net)

Method. Ridge minimises MSE + (λ/2)||w||² with our gradient descent; Lasso and Elastic Net use proximal gradient descent (a gradient step on the smooth part, then soft-thresholding by lr·λ·α, with α the L1 share: 0.25, 0.5, 0.9). All start from Phase 3's weights and its 589-column matrix, with step 0.9/(L + λ(1-α)) (the L2 term adds curvature; the plain 0.9/L step diverges for λ above about 4), λ log-spaced from 1e-4 to 10 (26 values; 16 for Elastic Net) and warm starts from the larger λ. All 100 fits converged. Phase 3 diagnosed bias, so we expected the optimum at small λ: all three methods peak at λ ≤ 0.07 and decay beyond it (figure).

Fair comparison and choice. Same grid, features, validation set and rule for the three methods: the largest λ whose validation R² is within 0.005 of that method's own best, which prefers the most compressed model that is still equally good.

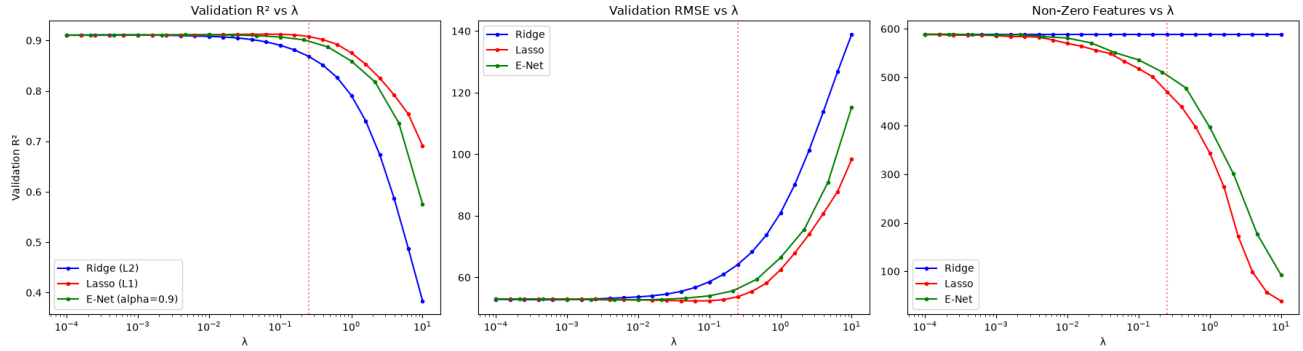
method	best R² (at λ)	chosen λ	validation R²	RMSE	features kept
Ridge (L2)	0.9112 (0.0006)	0.01585	0.9069	54.03	589 of 589
Lasso (L1)	0.9125 (0.063)	0.2512	0.9079	53.72	**470** of 589
Elastic Net (α = 0.9)	0.9114 (0.0046)	0.1	0.9069	54.02	536 of 589
unregularised (Phase 3)	-	0	0.9104	53.00	589 of 589

The differences between methods (about 0.001) are smaller than the fold-to-fold CV spread (±0.006), so none is more accurate than the others and none beats the unregularised Phase 3 model; Lasso is the only one that selects features and is carried forward: 119 features (20.2%) are removed for a cost of 0.0025 R².

Verdict for every original column. For each of the 14 input columns we removed all features derived from it (dummies, powers, interactions) and refitted the chosen Lasso. **Useful** if ΔR² ≥ 0.001; otherwise **Redundant** if |r| ≥ 0.85 with another column, else **Uninformative**.

column	kept / derived	ΔR²	closest partner	verdict
hr	455 / 552	0.5752	hum (0.28)	Useful
yr	22 / 24	0.0306	instant (0.87)	Useful
workingday	22 / 24	0.0190	holiday (0.25)	Useful
mnth	230 / 264	0.0113	season (0.97)	Useful
weathersit	38 / 48	0.0103	hum (0.40)	Useful
temp	24 / 26	0.0079	atemp (0.98)	Useful
hum	22 / 26	0.0052	weathersit (0.40)	Useful
weekday	100 / 144	0.0029	holiday (0.11)	Useful
instant	0 / 0	0.0000	dteday (1.00)	Redundant
dteday	1 / 1	0.0004	instant (1.00)	Redundant
season	1 / 3	-0.0000	mnth (0.97)	Redundant
atemp	1 / 1	0.0002	temp (0.98)	Redundant
holiday	0 / 1	-0.0000	workingday (0.25)	Uninformative
windspeed	2 / 4	0.0005	hum (0.32)	Uninformative

hr dominates; yr, workingday, mnth, weathersit, temp, hum and weekday follow. instant and dteday are perfectly correlated (instant is only a counter), season repeats mnth and atemp repeats temp. holiday has all its terms zeroed (workingday already encodes it) and windspeed adds 0.0005. yr has |r| = 0.87 with instant but is useful because hr x yr carries the growth over time, which is why the rule looks at ΔR^2 first. **Chronological check:** the chosen Lasso scores 0.8938 (RMSE 70.91) against 0.9063 for the unregularised model, so pruning costs 0.0025 on the seeded split and 0.0125 out of time.



Phase 4: validation R^2, RMSE and number of non-zero features against λ for Ridge, Lasso and Elastic Net (dotted line: chosen Lasso λ).

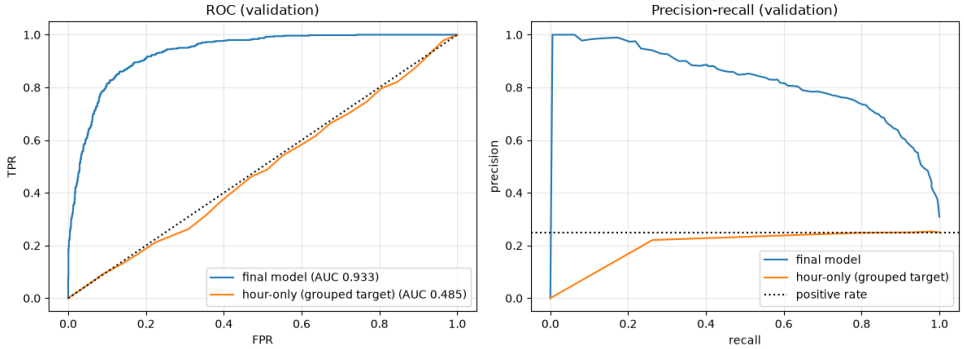
Phase 5 - Logistic regression

Target. One global threshold (the training 75th percentile of cnt, 283 bikes) would call every rush hour 'high' and every night hour 'low': 63.8% of its positives fall in the 7 commute hours (an even spread would be 29.2%), and a logistic model that only knows the hour reaches AUC 0.850 on it. We therefore define **high = cnt above the training 75th percentile of the same (hour, workingday) group** (48 groups, computed from training rows only; for example 651 bikes at 8 a.m. on working days and 144 on non-working days): 24.4% positives in train and 25.0% in validation. The same hour-only model now scores AUC 0.485, i.e. guessing, so the classifier has to use weather, season and trend.

Model and tuning. Logistic regression with the same gradient descent: a numerically stable log-loss plus $(L2/2)||w||^2$ (gradient checked against finite differences, relative error below 1e-8), step 1/L with $L = \lambda\max(X^TX/n)/4 + L2$, on exactly the 470 features Phase 4 kept (hash-verified). We compared 12 settings (L2 in {1, 0.1, 0.01, 1e-3, 1e-4, 0} x class weights off/on) by AUC and by F1 at each model's own best threshold: AUC ranges from 0.883 to 0.936 and class weights never improve it (they only move the best threshold from about 0.4 to about 0.65). Four settings are within 0.005 F1 of the best; we take the most regularised, **L2 = 0.01 without class weights** (105 iterations). Because 75% of the hours are negative, accuracy alone is misleading (always answering 'not high' scores 0.750), so we report F1 and ROC-AUC and choose the decision threshold on validation F1: **0.38**. Tuning and reporting on the same validation set makes the tuned numbers slightly optimistic.

model	accuracy	precision	recall	F1	ROC-AUC
logistic, 470 features, threshold 0.38	0.8788	0.7322	0.8128	**0.7704**	**0.9331**
same model, threshold 0.5	0.8747	0.7845	0.6881	0.7331	0.9331
baseline: always 'not high'	0.7498	0	0	0	0.5000
baseline: hour-only logistic (best-F1 threshold)	0.2718	0.2530	0.9780	0.4020	0.4850

The model finds 443 of the 545 high validation hours (missing 102) and wrongly flags 162 others. F1 stays near 0.77 although the AUC is 0.93: the label is a sharp cut through a noisy count, so hours just below or just above their limit are nearly indistinguishable (our explanation, not tested separately). On the chronological split the AUC is 0.8125; there 64% of the hours exceed limits learned from earlier months (demand grew in 2012), so only the AUC is comparable with the seeded split.



Phase 5: ROC and precision-recall curves on the validation set, final model against the hour-only baseline.

Test predictions and recommended regression model

The recommended regression model is the **unregularised Phase 3 interaction model (R9, 589 features)**, because nothing beat it: Ridge, Lasso and Elastic Net reach 0.9069, 0.9079 and 0.9069 on validation (against 0.9104), and the Lasso 0.8938 against 0.9063 on the chronological split. We refit it on all 10,886 labelled rows (allowed if justified): the day-grouped 5-fold CV refits the whole recipe (preprocessing, expansion, gradient descent) in every fold and gives R^2 **0.9148 raw and 0.9209 after flooring at 0** (RMSE 52.68 and 50.75), and the test days are interleaved with the training days, so 25% more labelled days help. Predictions are floored at 0 because counts cannot be negative (33 of the 574 test predictions, 5.7%, were negative; the lowest was -105). The final fit (momentum, 1,448 iterations) matches the closed-form training loss to 4.7e-6 (relative) and is bit-identical on a second run. *sample_submission.csv* has 574 rows in *test.csv* order; the mean prediction (197.4) is close to the training mean (191.6) and every hourly mean prediction is within 25.4 bikes of the hourly training mean.

Pipeline retrospective

phase	model	features	validation (seeded split)	chronological split*
1	linear, own gradient descent	54	R^2 0.6781, RMSE 100.44	R^2 0.6364
2	powers of temp, hum, windspeed, days (degree 3)	62	R^2 0.6858, RMSE 99.24	R^2 0.5099
3	+ hour interactions, no time powers (R9)	589	R^2 0.9104, RMSE 53.00	R^2 0.9063
4	Lasso, $\lambda = 0.2512$	470	R^2 0.9079, RMSE 53.72	R^2 0.8938
5	logistic regression on the Phase 4 features	470	AUC 0.9331, F1 0.7704, accuracy 0.8788	AUC 0.8125

**Phases 1 and 2 were not evaluated chronologically in their own phase; their chronological values are the R0 and R1 rows of the Phase 3 ladder.*

The large gain came from the **structure** of the model (hour interactions: +0.225 of R^2), not from curvature (+0.008) or regularization (-0.0025); the time-ordered split exposed the polynomial in days_since_start that random splits hid, and the out-of-time number is always a little lower than the random-split one. The hand-off between phases (Phase 1 weights into Phase 2, Phase 2 into Phase 3's ladder, Phase 3's model into Phase 4, Phase 4's 470 features into Phase 5, Phase 3's recipe into the submission) is hash-checked at every step. **Limits.** Choices made on the seeded validation set, as the project rules prescribe (Phase 2's degree, Phase 4's λ, Phase 5's L2 and threshold), make those validation numbers slightly optimistic; the hum = 0 rows and the 58 missing hours are kept as they are; the cut-off explanation of the Phase 5 F1 is untested.